Apply filters to SQL queries

Project description

Using SQL filters like AND, OR, and NOT to retrieve records fromdatasets such as `log_in_attempts` and `employees` tables from `organization` database to investigate potential security issues.

Retrieve after hours failed login attempts

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |       0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   |       0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   |       0 |
|       69 | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  |       0 |
|       82 | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  |       0 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |       0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  |       0 |
|      104 | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  |       0 |
|      107 | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 |       0 |
|      111 | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   |       0 |
|      127 | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  |       0 |
|      131 | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 |       0 |
|      155 | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 |       0 |
|      160 | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  |       0 |
|      199 | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
19 rows in set (0.277 sec)

MariaDB [organization]>
```

- To filter failed login attempts and also login after 18:00 which is after business hour, we will filter the database using SQL queris.
- We are selecting all the coulumns using `SELECT *` query from log_in_attempts table using `FROM log_in attempts` query which can be seen in line 1 and 2.
- Then we filter it using WHERE clause and operators. Line 3 shows `WHERE login_time > '18:00' AND success = FALSE` . Which means we are filtering using conditions like the login_time column should be after 18:00 and the success rate of login should be failed. By using AND operator we are asking both the conditions should be satisfied.
- And we got 19 rows of output by satisfying the condition.

Retrieve login attempts on specific dates

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|      184 | atevitsk | 2022-05-08 | 03:09:48   | CAN     | 192.168.33.70   |       0 |
|      186 | bisles   | 2022-05-09 | 04:29:17   | USA     | 192.168.40.72   |       0 |
|      187 | arusso   | 2022-05-09 | 00:36:26   | MEX     | 192.168.77.137  |       0 |
|      189 | nmason   | 2022-05-08 | 05:37:24   | CANADA  | 192.168.168.117 |       1 |
|      190 | jsoto    | 2022-05-09 | 05:09:21   | USA     | 192.168.25.60   |       0 |
|      191 | cjackson | 2022-05-08 | 06:46:07   | CANADA  | 192.168.7.187   |       0 |
|      193 | lrodriqu | 2022-05-08 | 07:11:29   | US      | 192.168.125.240 |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
75 rows in set (0.001 sec)

MariaDB [organization]>
```

- There was a suspicious event occurred on 2022-05-09, so we are reviewing login attempts on this day and the day before using SQL queries from log_in_attempts database.
- In line 1 and 2, we are selecting all columns from log_in_attempts table using SQL queries `SELECT * FROM log_in_attempts`
- In line 3, we are applying filter using OR operator, where either of the condition should be satisfied. The filter using SQL queries is `WHERE login_date = '2022-05-09' OR login_date = '2022-05-08'` , here we are filtering the logn_date so either any one of the date 2022-05-08 or 2022-05-08 should be applied.
- And we got 75 rows of output by satisfying the condition and the output screenshot shows the cropped image of first few and last few lines.

Retrieve login attempts outside of Mexico

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        5 | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  |       0 |
|        7 | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 |       1 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|       10 | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 |       0 |
```

```
|      194 | jclark   | 2022-05-12 | 14:11:04   | CAN     | 192.168.197.247 |       0 |
|      195 | alevitsk | 2022-05-11 | 06:59:13   | CANADA  | 192.168.236.78  |       1 |
|      196 | acook    | 2022-05-10 | 09:56:48   | CAN     | 192.168.52.90   |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
|      200 | jclark   | 2022-05-12 | 01:11:45   | CANADA  | 192.168.91.103  |       1 |
+----------+----------+------------+------------+---------+-----------------+---------+
144 rows in set (0.001 sec)
```

- The team has determined that the suspicious activity of login attempts is didn't originate from Mexico. So we are applying filter to identify all login attempts outside of Mexico. And also in the country column in the log_in_attempts table database has MEX and MEXICO as a keyword for Mexico, so we also apply filter to consider both keywords.
- In line 1 and 2, we are selecting all the columns from log_in_attempts table using SQL query `SELECT * FROM log_in_attempts` .
- We are applying condition using NOT operator, so it returns everything except the condition mentioned in the line 3. `WHERE NOT country LIKE 'MEX%'` , here NOT operator will return everything except this condition LIKE 'MEX%' . Here we used LIKE operator by introducing % wildcard because it will consider all the data starting from MEX keyword.
- And the output is filtered as expected and we got data of 144 rows satisfying the condition. The output image is cropped by first part and the last part of the output.

Retrieve employees in Marketing

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
+-------------+-------------+----------+------------+----------+
| employee_id | device_id   | username | department | office   |
+-------------+-------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
|        1088 | k865l965m233 | rgosh    | Marketing  | East-157 |
|        1103 | NULL         | randerss | Marketing  | East-460 |
|        1156 | a184b775c707 | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339 | cwilliam | Marketing  | East-216 |
+-------------+-------------+----------+------------+----------+
7 rows in set (0.040 sec)

MariaDB [organization]>
```

- Team wants to perform security updates on specific employee machines in the Marketing department of East building. So using SQL query we filter the data required from the employees table from the organization database.
- Line 1 and 2, tells that to select all the columns in the employees table using SQL query `SELECT * FROM employees` .
- We are using AND operator to satisfy both the condition like selecting only Marketing department and in the East building office. Since East buliding has multiple values like East-170, East-460 and so on, we are using LIKE operator with % wildcard.

- In line 3, WHERE department = ‘Marketing’ AND office LIKE ‘East%’ is the query we used. Here only Marketing department will be filtered from department column with East building from office column.
- And we got 7 rows of output by satisfying all the conditions we applied.

Retrieve employees in Finance or Sales

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
+-------------+-------------+----------+------------+-------------+
| employee_id | device_id   | username | department | office      |
+-------------+-------------+----------+------------+-------------+
|        1003 | d394e816f943 | sgilmore | Finance    | South-153   |
|        1007 | h174i497j413 | wjaffrey | Finance    | North-406   |
|        1008 | i858j583k571 | abernard | Finance    | South-170   |
|        1009 | NULL         | lrodriqu | Sales      | South-134   |
|        1010 | k242l212m542 | jlansky  | Finance    | South-109   |
|        1011 | l748m120n401 | drosas   | Sales      | South-292   |
|        1186 | e281f433g404 | sacosta  | Sales      | North-460   |
|        1187 | f963g637h851 | bbode    | Finance    | East-351    |
|        1188 | g164h566i795 | noshiro  | Finance    | West-252    |
|        1195 | n516o853p957 | orainier | Finance    | East-346    |
+-------------+-------------+----------+------------+-------------+
71 rows in set (0.001 sec)

MariaDB [organization]>
```

- Our team now needs to perform a different security update on machines for employees in the Sales and Finance departments. By using filters in SQL to create a query so that it can identifies all employees in the Sales or Finance departments.
- In line 1 and 2, we are selecting all the columns from the employees table from organization database using SQL query SELECT * FROM employees .
- In line 3 we are using WHERE department = ‘Finance’ OR department = ‘Sales’ ,here we used OR operator because we need either Finance department or Sales department from department column from employees table. By using OR operator if the department columns value is either Finance or Sales it will return the data.
- And we got 71 rows of data output satisfying the condition. Cropped image of first part and the last part of the output is attached.

Retrieve all employees not in IT

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
+-------------+-------------+----------+-----------------+-------------+
| employee_id | device_id   | username | department      | office      |
+-------------+-------------+----------+-----------------+-------------+
|        1000 | a320b137c219 | elarson  | Marketing       | East-170    |
|        1001 | b239c825d303 | bmoreno  | Marketing       | Central-276 |
|        1002 | c116d593e558 | tshah    | Human Resources | North-434   |
|        1003 | d394e816f943 | sgilmore | Finance         | South-153   |
|        1004 | e218f877g788 | eraab    | Human Resources | South-127   |
|        1005 | f551g340h864 | gesparza | Human Resources | South-366   |
+-------------+-------------+----------+-----------------+-------------+
|        1191 | NULL         | shakimi  | Marketing       | Central-366 |
|        1194 | m340n287o441 | zwarren  | Human Resources | West-212    |
|        1195 | n516o853p957 | orainier | Finance         | East-346    |
|        1198 | q308r573s459 | jmartine | Marketing       | South-117   |
|        1199 | r520s571t459 | areyes   | Human Resources | East-100    |
+-------------+-------------+----------+-----------------+-------------+
161 rows in set (0.001 sec)

MariaDB [organization]>
```

- Now team needs to make one more update to employee machines. The employees who are in the Information Technology department already had this update, but employees in all other departments need it. By filtering using SQL query we will identify all employees not in the IT department.
- Line 1 and 2, shows we are selecting all the columns in the employees table using SQL query `SELECT *` `FROM employees` .
- In line 3 we applying filter using NOT operator, NOT operator will return every value except the mentioned condition. `WHERE NOT department = 'Information Technology'` will return all the departments in the department column from the employees table except Information Technology department.
- And we got expected output satisfying all the condition. The image has cropped image of first part and the last part of the output.

Summary

When there is large amount of data in the database, filtering the data that we required to make security related decisions so that we can keep the system secure of the organization. Filtering using SQL operators such as AND, OR, and NOT operator and also combining them with the other operators such as LIKE will make the work easier which we have done in the above scenario.